BIO102 Help Session - 1

The purple grain wheat variety called Saravtoskaya-29 (S29) is an isogenic variety (i.e., all the plants are clones of each other). In a study conducted in 2017 in Turkey, a large area was sown with S29 variety and then harvested. The mean wheat yield per plant was 103 grams with a standard deviation of 8 grams. Calculate (a) the components of total phenotypic variation and (b) the broad sense heritability for wheat yield per plant in the S29 variety. (2 marks)

1. Mendelian traits are characters with low environmental variance and simple genotype-phenotype mapping. Explain this statement. (2 marks)

Monohybrid Cross

In a certain species of plants, purple flowers are dominant to white. When a particular purple-flowered plant is self-fertilized, the progeny are 28 purple-flowered and 11-white flowered.

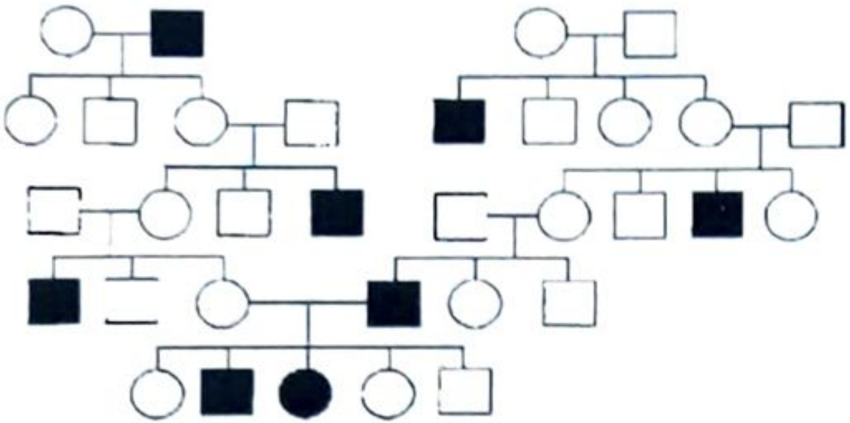
(a) What is the genotype of the plant that was selfed?

(b) What proportion of the purple-flowered progeny is expected to be true-breeding (pure-breeding, also called “breed true”)? (c) What proportion of the white-flowered progeny will breed true?

Dominance relationships

4. Explain the different types of dominance relationships that are possible across alleles at a locus. (2 marks)

5. Consider the following pedigree-



a. What is the most probable mode of inheritance? (Autosomal or X linked/ Dominant or recessive). With reference to the pedigree, point out and explain the strongest evidence in favour of your answer. (1+ 2 marks)

b. If the individual V-2 marries a homozygous normal person, what is the probability that their first child will be heterozygous for the condition? (2 marks)

Dihybrid cross

Q. What gametes can be formed by an individual organism of genotype Aa? Of genotype Bb? Of genotype Aa Bb?

Q. How many different gametes can be formed by an organism with genotype AA Bb Cc Dd Ee and, in general, by an organism that is heterozygous for m genes and homozygous for n genes?

Dihybrid cross

Suppose that in a plant species, spiny pods (S) are dominant to smooth (s). Additionally, Purple flowers (P) are dominant to white flowers (p). Consider pod shape and flower color together.

3. Consider the plant species mentioned in question (2). Suppose that a pure breeding Spiny, purple plant is crossed with a pure breeding smooth, white plant. (a) What will be the phenotype of F1 and F2? (b) What will be the phenotypes of the progeny from a cross between the F1 and the Purple, spiny parent? (c) What will be the phenotypes of the progeny from a cross between the F1 and the smooth, white parent? (d) In the F2 generation, what proportion of the progeny will be homozygous for dominant alleles at both loci?

7. Differentiate between segregation and independent assortment. (2 marks)

BIO 102: Mid Semester Examination 1

Total Marks: 25

Registration Number:

Name:

3. Assume that in a dihybrid cross starting from pure breeding parents, the F2 shows a 13:3 ratio for two phenotypes. If we take the F1 individual from the above cross and subject it to a test cross, what is the expected phenotypic ratio? Explain. (3 marks)

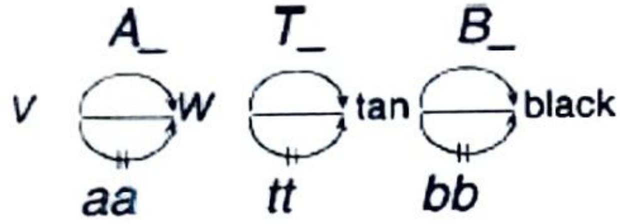
BIO 102: Mid Semester Examination 1
Total Marks: 25

Registration Number:
Name:

6. Observe the particular set of Drosophila crosses using pure breeding individuals of four eye colours- Orange-1; Orange- 2; Pink and Wild type. Write down the genotype of the four types of eye colour individuals. (3 marks)

Cross	**F1 Phenotype**
Wild-type X Orange-1	All wild-type
Wild-type X Orange-2	All wild-type
Orange-1 X Orange-2	All wild-type
Orange-2 X pink	All Orange-2
(F1 of (Orange-1 X Orange-2)) X pink	¼ Orange-2; ¼ pink; ¼ orange-1; ¼ wild-type

8. Assume that in rabbits there are three different independently assorting autosomal loci that affect coat color. A colorless-pigment precursor V is converted to a colorless precursor W by action of the A allele. W is converted to tan pigment by action of the T allele, and the tan pigment is converted to black pigment by action of the B allele.



The homozygous recessive condition at each locus results in loss of enzyme activity for the reaction controlled by that gene. Write down the expected phenotypic ratios for a cross of trihybrid bunnies, AaTtBb X AaTtBb. (3 marks)

BIO 102: Mid Semester Examination 1

Total Marks: 25

Registration Number:

Name:

9. The autosomal genes Curly (Cy) and Plum (Pm) in Drosophila are independently segregating, phenotypically dominant mutations that are recessive lethals. In other words, they have a distinct phenotype as heterozygotes but are lethal as homozygotes. When Curly and Plum are separately crossed to wild flies and the F1 mutant progeny are intercrossed, what will the resulting phenotypic ratio be? (3 marks)

3.4 Phenylketonuria is a recessive inborn error of metabolism of the amino acid phenylalanine that results in severe mental retardation of affected children. The female II-3 (red circle) in the pedigree shown here is affected. If persons III-1 and III-2 (they are first cousins) mate, what is the probability that their offspring will be affected? (Assume that persons II-1 and II-5 are homozygous for the normal allele.)